

# **Assignment- 01**

1. Write a code for the clock divider module. Divide the on board 100Mhz clock to 50Mhz, 25Mhz and 50Hz. (Hint: Try to use counter variable to do this)

2. Write a code to implement a basic 2bit adder on the FPGA. Map the inputs taken respectively to the physical switches on the board. Take the help of this [Artix-7 reference manual](#) for I/O planning.

[For the next three questions use the previously designed clock divider module if needed]

3. Write a code to implement a proper Up-Down counter on the FPGA. Use the switch J15 as the up/down counting controller. The result should be reflected on both seven segment display and the LED's. Take the help of this [Artix-7 reference manual](#) for I/O planning.

4. Write a code to display a stable white screen on the monitor.

5. Now display your name on the monitor. (Firstly draw the pixel diagram i.e what pixels to be in what colour and then write the code).